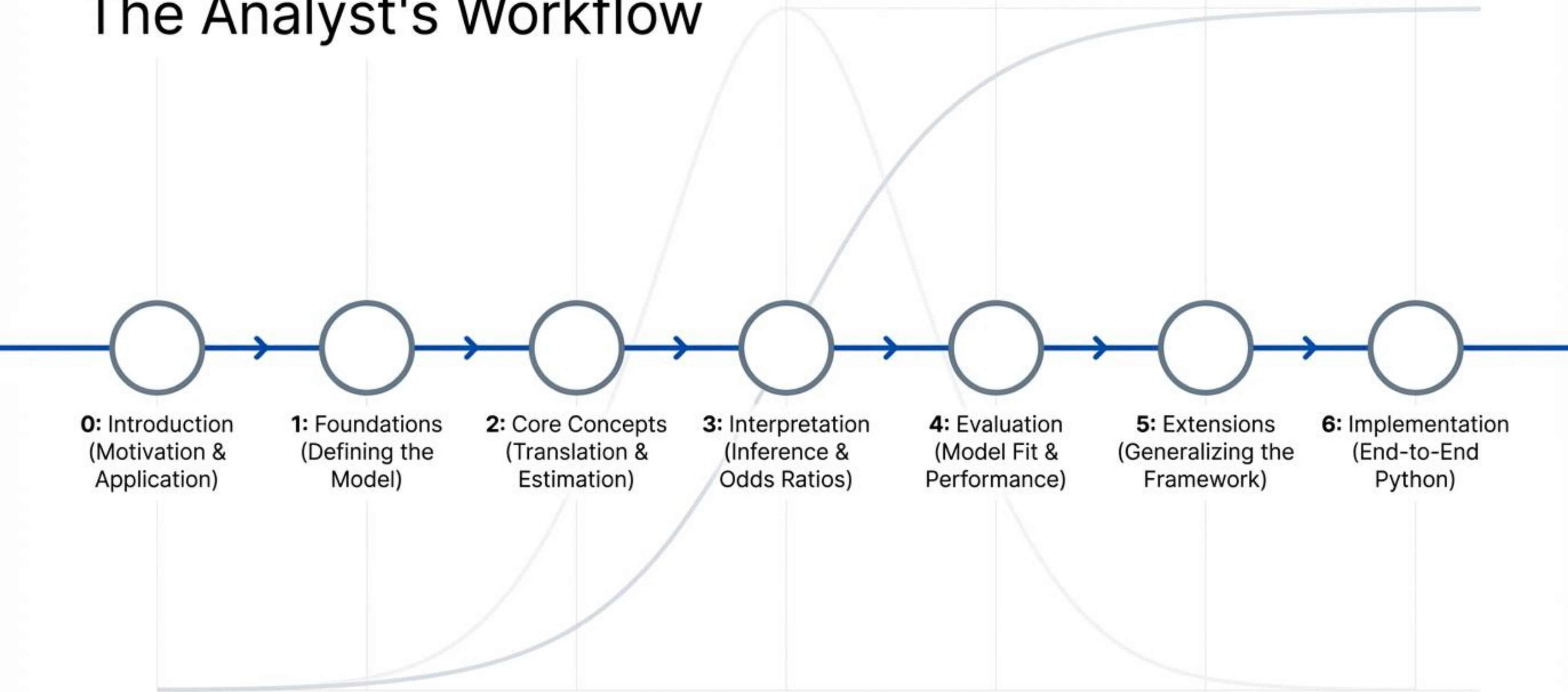


Logistic Regression

The Analyst's Workflow



Here we have the
independent variables

and here the **dependent variable** with 0 and 1.



Age	Gender	Smoker status	Disease
22	female	Non-smoker	1
25	female	Smoker	1
18	male	Smoker	0
45	male	Non-smoker	0
12	female	Smoker	0
43	male	Smoker	1
23	male	Smoker	0
33	male	Smoker	1
...	...	...	...

0. Introduction to Logistic Regression

Purpose and Motivation

Mathematical Purpose

Definition: A probabilistic model used to describe the relationship between one or more continuous/categorical predictor variables and a discrete, binary categorical outcome.

The Problem: Real-world phenomena rarely scale infinitely; they resolve into discrete states (Success/Failure, Yes/No).

Typical Applications - Classification Problems



Political Science: Predicting voter choice (e.g., Democrat vs. Republican) based on income and ideology.



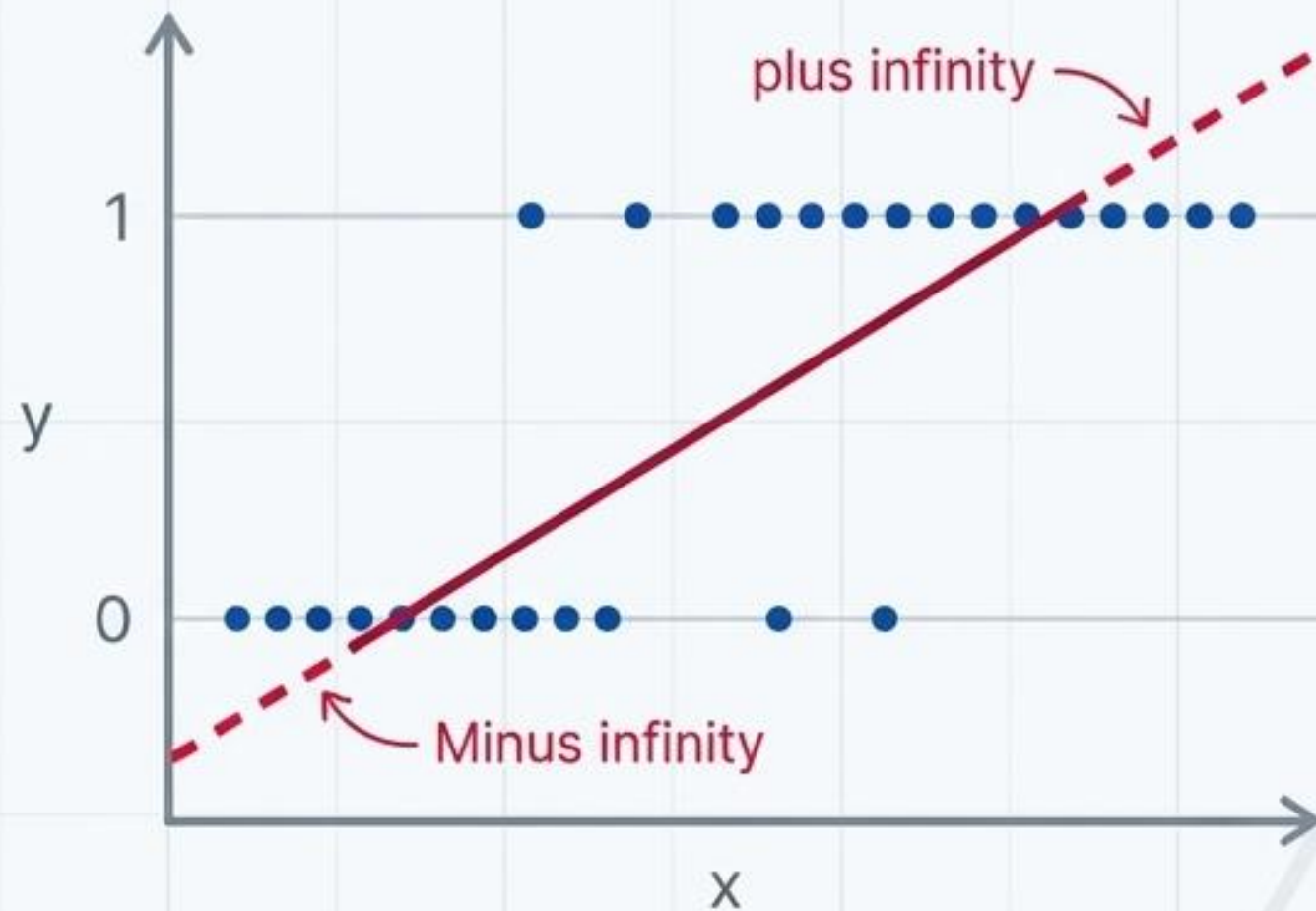
Medicine: Determining disease susceptibility (e.g., Diseased vs. Healthy) given age, gender, and treatment.



Business: Predicting consumer behavior (e.g., Buy vs. Do Not Buy) from demographic data.

1.1 Defining Binary Logistic Regression

Why Linear Regression Fails for Binary Data

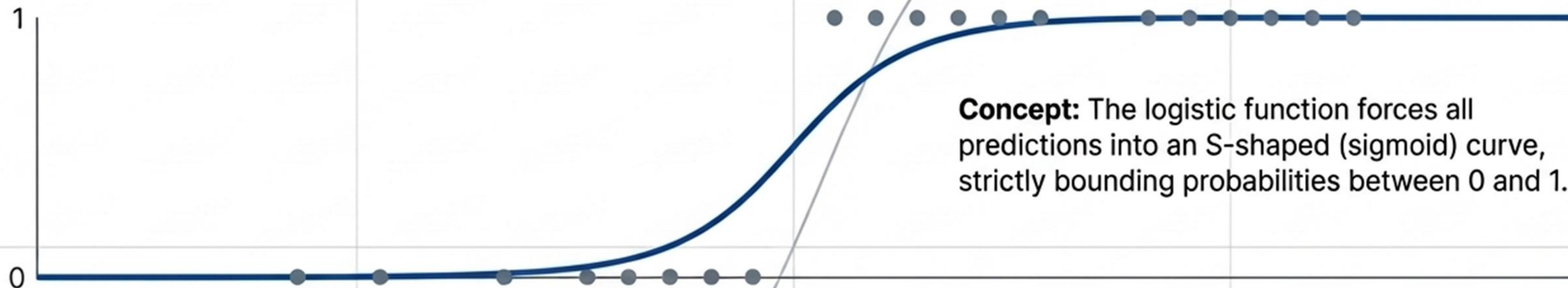


The Overshoot Problem: The Linear Probability Model $E(y) = \alpha + \beta x$ allows probabilities to fall below 0 or rise above 1, violating the fundamental axioms of probability.

Dimension	OLS (Linear)	Logistic Regression
Outcome Type	Continuous	Binary
Boundaries	$-\infty$ to ∞	0 to 1
Error Distribution	Normal	Binomial
Interpretation	Unit Change in Y	Multiplicative Odds Ratio

1.2 The Logistic Model

The Logistic Function and the S-Shaped Curve



Inter	Translation	
Logistic Function	$p = \frac{1}{1 + e^{-(\alpha + \beta x)}}$	The probability of success (p) equals 1 divided by 1 plus the mathematical constant e raised to the negative power of the linear regression equation.
Logit (Linear Equation)	$\ln\left(\frac{p}{1-p}\right) = \alpha + \beta x$	The natural log of the odds (logit) is a linear function of the predictors.

Interpretation in Probabilities: The S-shape implies that the rate of change in probability varies. A one-unit change in x has the greatest impact on probability when $p = 0.50$ (the steepest part of the curve).

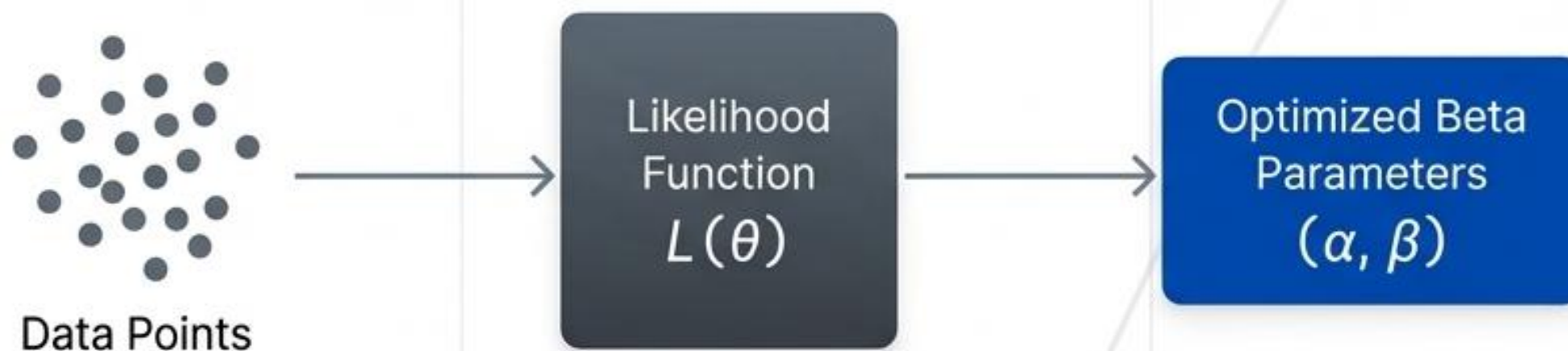
2.1 Probabilities, Odds, and Logits

The Translation Triad

The Scale of Outcomes		
Probability (p)	Odds	Logit (Log-Odds)
$\frac{\text{Number of successes}}{\text{Total trials}}$	$\frac{p}{1 - p}$	$\ln\left(\frac{p}{1 - p}\right)$
• Range: Bounded strictly between 0 and 1. • Meaning: The raw likelihood of an event occurring.	• Range: Bounded at 0, extends to $+\infty$. • Meaning: The ratio of success to failure (e.g., 3:1 odds means 3 successes for every 1 failure).	• Range: Unbounded ($-\infty$ to $+\infty$). • Meaning: The natural log of the odds. This transformation creates a linear, continuous variable suitable for regression modeling.

2.2 Model Estimation

Maximum Likelihood Estimation (MLE)



An iterative algorithm tests different coefficients to maximize the log-likelihood function. It stops when it finds the peak probability of generating the exact dataset observed.

OLS (Least Squares)

Minimizes the sum of squared errors.

MLE (Maximum Likelihood)

Identifies the parameter values that make the observed data most mathematically probable.

The Null Model (L_0)

- The baseline model containing only an intercept (α), assuming no predictors have an effect.
- Model Comparison: We evaluate our fitted model (L_1) against the null model (L_0) to prove our predictors actually improve predictive accuracy.

2.3 Assumptions for Statistical Inference

The Structural Integrity of the Model

Statistical Inference

Independence of Observations

Cases must not be correlated with each other (e.g., no repeated measures on the same subject without mixed modeling).

Linearity in the Logit

Continuous predictors must have a linear relationship with the log-odds (logit) of the outcome, despite non-linear probability.

Absence of Multicollinearity

Predictor variables must not be too highly correlated, which inflates standard errors and destabilizes coefficient estimates.

Adequate Sample Size

MLE requires large sample sizes. Small samples or perfectly separated data cause parameter estimates to approach infinity.

3.1 Interpreting Model Coefficients

From Logits to Real-World Effects

Predictor	Coefficient (B)	S.E.
Victim: White	2.404	0.601
Prior Arrests	0.510	0.120

Coefficient (B) & Direction

B represents the linear change in the log-odds for a 1-unit increase in X.

Exponentiating Coefficients (e^B)

To translate abstract log-odds into an intuitive Odds Ratio, take the antilog (e^B).

Interpretation & Example (Death Penalty Verdicts)

- $e^{2.404} = 11.06$. The odds of receiving the death penalty are 11 times higher when the victim is white compared to black.
- For each additional prior arrest ($e^{0.510} \approx 1.67$), the odds increase by a factor of 1.67.

3.2 Statistical Inference

Testing Significance and Fit

The Wald Test

Testing Individual Effects

- **Purpose:** Tests the null hypothesis ($H_0: \beta = 0$) for a single independent predictor.
- **Math:** The square of the ratio of the estimate to its standard error ($z = \hat{\beta} / se$).
- **Note:** Can be unreliable in small samples; serves as a quick check for individual variable impact.

Likelihood-Ratio Test

Model Assessment (Chi-Squared)

- **Purpose:** The gold standard for comparing nested models. Tests if adding a set of variables significantly improves model fit over a simpler model.
- **Math:** Compares the maximized log-likelihoods ($-2 \log L$) between the full and reduced models.

P-values: Provide the statistical significance for both tests. A small p-value (<0.05) **rejects** the null hypothesis, mathematically **confirming** the predictor's effect.

4.1 & 4.2 Model Evaluation: Fit and Accuracy

The Diagnostic Dashboard

Model Fit: Pseudo R-Squared Measures

Cox & Snell

0.315



Compares likelihoods but cannot reach a theoretical maximum of 1.0.

Nagelkerke

0.462



Adjusts Cox & Snell so the scale properly spans 0 to 1.

McFadden

0.248



Compares the log-likelihood of the full model versus the intercept-only model.

Unlike OLS, logistic regression has no true “variance explained” metric. Pseudo R-Squared approximates effect size.

Predictive Performance: Classification Matrix

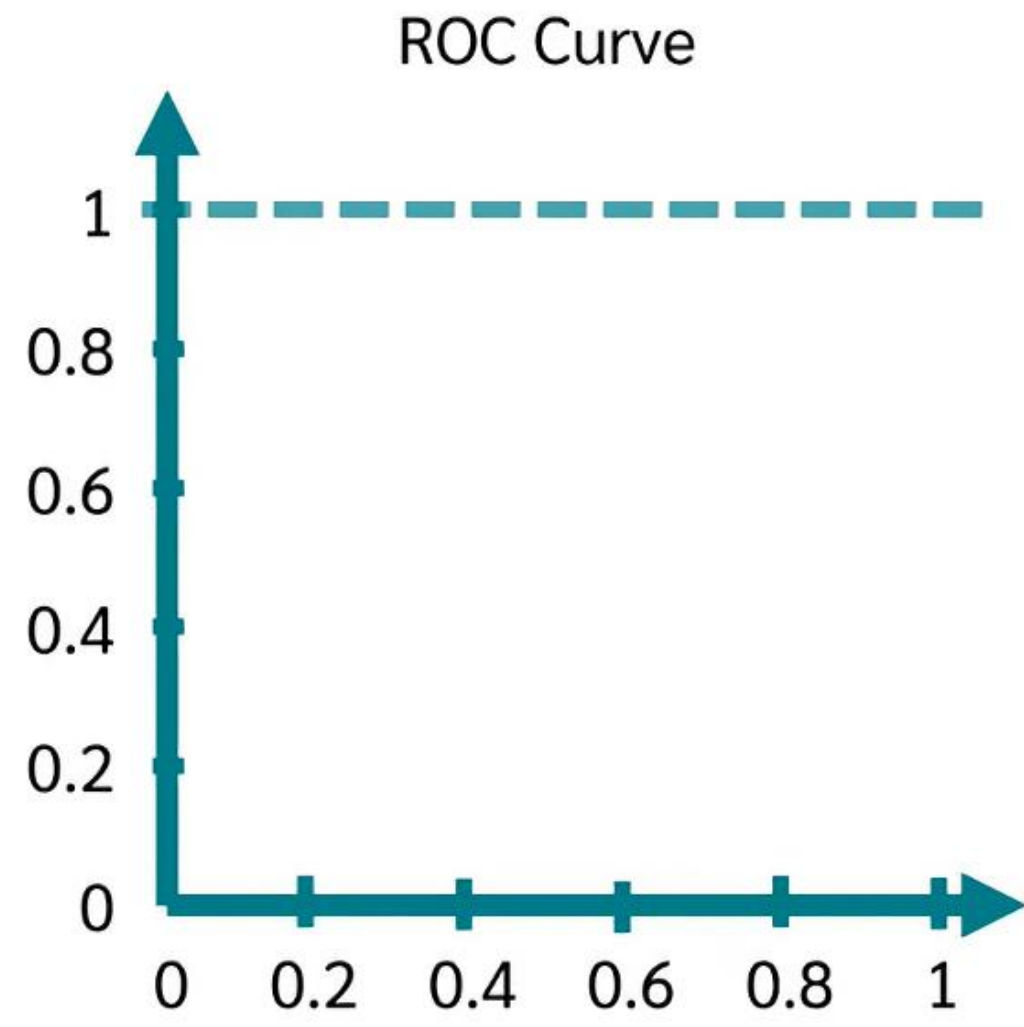
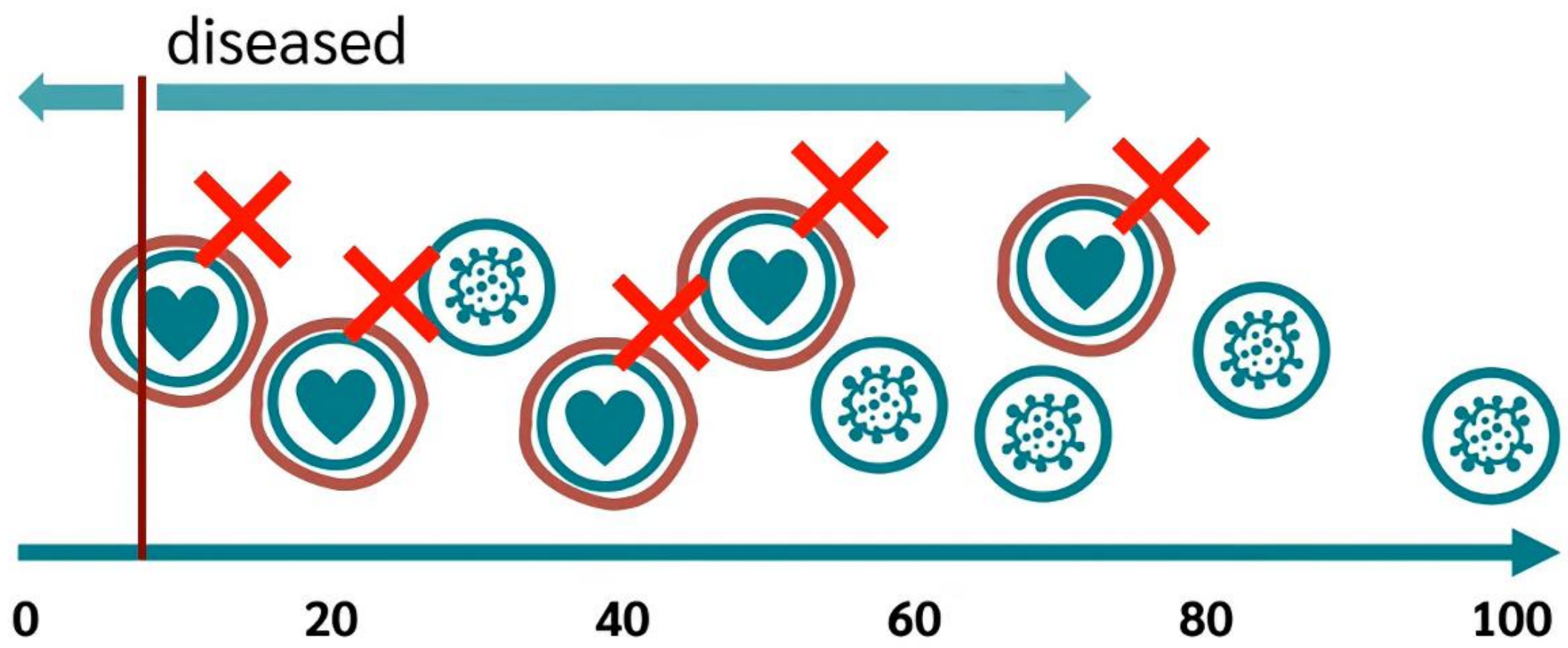
		Predicted Outcome	
		Positive (1)	Negative (0)
Observed Outcome	Positive (1)	TP 350	FN 50
	Negative (0)	FP 80	TN 520

True Positives points to the TP cell (350).

True Negatives points to the TN cell (520).

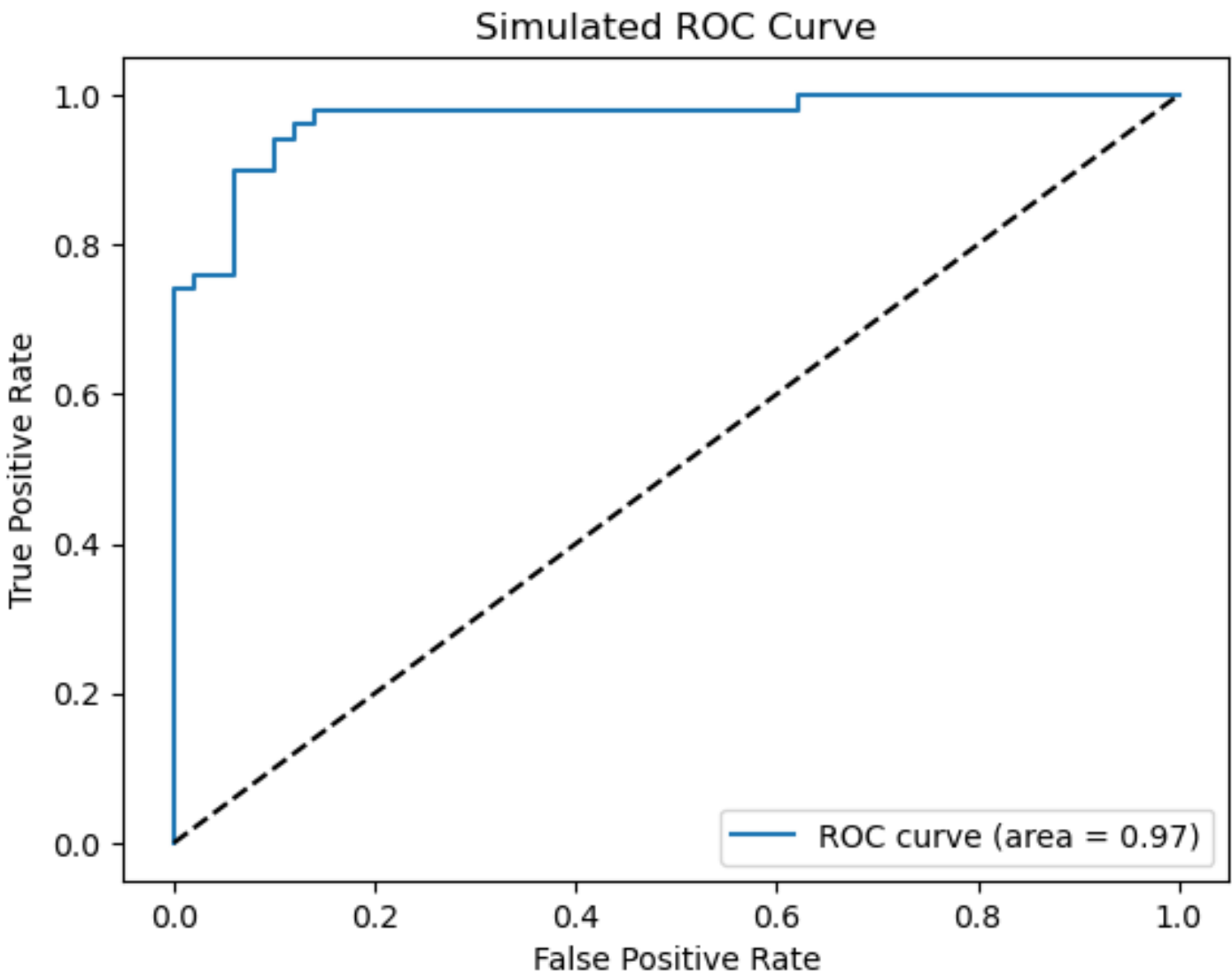
Classification Tables: Cross-tabulates observed binary outcomes against predicted outcomes using a probability threshold (usually 0.50).

Hit Rate (Accuracy): The percentage of total cases correctly classified by the model. $(TP + TN) / \text{Total Cases} = (350 + 520) / 1000 = 0.870$ (87.0%). ✓



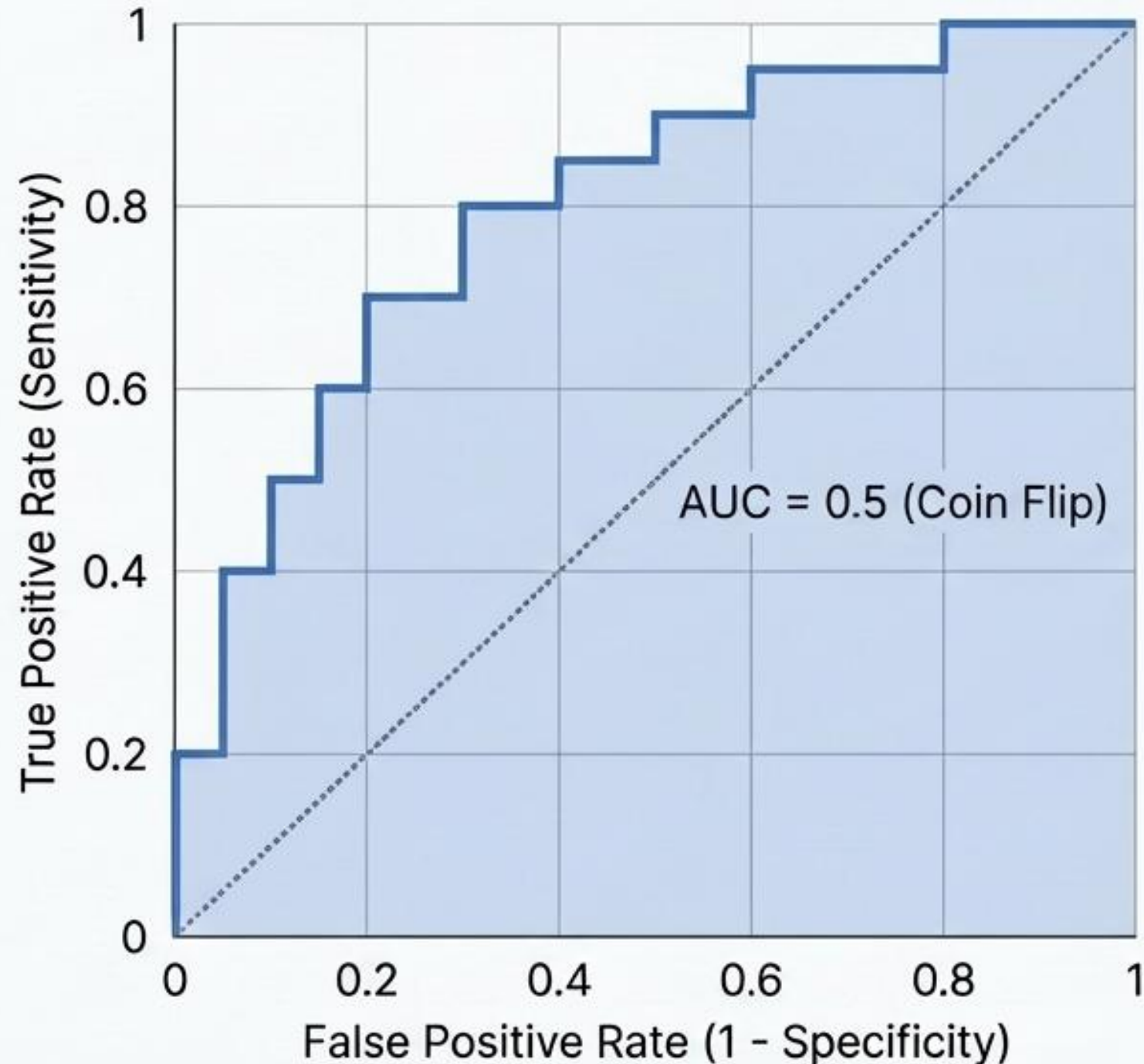
Data for plotting ROC curve

Thresho Id	Classified as "Positive"	Count (TP, FP)	TPR (Sensitivity)	FPR (1- Spec)	Resulting ROC Point
0.9	A	1 TP, 0 FP	0.2 (1/5)	0.0 (0/5)	(0.0,0.2)
0.8	A, B	2 TP, 0 FP	0.4 (2/5)	0.0 (0/5)	(0.0,0.4)
0.7	A, B, C	3 TP, 0 FP	0.6 (3/5)	0.0 (0/5)	(0.0,0.6)
0.6	A, B, C, D	3 TP, 1 FP	0.6 (3/5)	0.2 (1/5)	(0.2,0.6)
0.5	A, B, C, D, E	4 TP, 1 FP	0.8 (4/5)	0.2 (1/5)	(0.2,0.8)
0.3	A thru G	5 TP, 2 FP	1.0 (5/5)	0.4 (2/5)	(0.4,1.0)
0	All (A thru J)	5 TP, 5 FP	1.0 (5/5)	1.0 (5/5)	(1.0,1.0)



4.2 Predictive Performance

ROC Curves and the Geometry of AUC



Receiver Operating Characteristic (ROC):

- Plots TPR against FPR across every possible probability threshold.
- Escapes the limitation of relying on a single arbitrary cutoff (like $p=0.5$).

Area Under the Curve (AUC):

- A single geometric metric summarizing model discrimination.
- AUC = 0.5: Useless model (diagonal coin-flip line).
- AUC = 1.0: Perfect predictive classification.
- Interpretation: The probability that the model will rank a randomly chosen positive case higher than a randomly chosen negative case.

5. Extensions and Generalizations

Expanding the Architecture Beyond Binary

Model Extensions Matrix	
Multiple Logistic Regression	- Structure: Binary outcome, but incorporates multiple independent predictors ($X_1, X_2, \dots, X_p$). - Mechanics: Isolates the partial effect of one variable while controlling for others.
Multinomial (Nominal) Logistic Regression	- Structure: Unordered categorical outcome (e.g., Car, Bus, Train). - Mechanics: Creates baseline-category logits. Pairs each category against a single reference baseline (e.g., Car vs. Train, Bus vs. Train).
Ordinal Logistic Regression	- Structure: Ordered categorical outcome (e.g., Disagree, Neutral, Agree). - Mechanics: Uses Cumulative Logits. Relies on the “Proportional Odds” assumption—assuming the effect of X is identical across all cumulative splits.

Quiz